Edelstein, Price, and Zerilli watched black holes pulsate in computer simulations and in pencil-and-paper calculations. What they thought they were seeing was gravitational radiation (ripples of spacetime curvature) bouncing around in the vicinity of a hole, trapped there by the hole's own spacetime curvature. The trapping was not complete; the ripples would gradually leak out of the hole's vicinity, and fly away. This was sort of cute, but not terribly interesting.

In autumn 1971, Bill Press, a new graduate student in my group, realized that the ripples of spacetime curvature bouncing around near a hole could be thought of as pulsations of the black hole itself. After all, as seen from outside its horizon, the hole consists of nothing but space-time curvature. The ripples of curvature were thus nothing more nor less than pulsations of the hole's curvature, and therefore pulsations of the hole itself.

This change of viewpoint had a huge impact. If we think of black holes as able to pulsate, then it is natural to ask whether there are any similarities between their pulsations and the pulsations ("ringing") of a bell, or the pulsations of a star. Before Press's insight, such questions weren't asked. Afterwards, such questions were obvious.

A bell and a star have natural frequencies at which they like to pulsate. (The bell's natural frequencies produce its pure ringing tone.) Are there similarly natural frequencies at which a black hole likes to pulsate? Yes, Press discovered, using computer simulations. This discovery triggered Chandrasekhar, together with Steven Detweiler (an intellectual great-grandson of Wheeler's), to embark on a project of cataloging all of a black hole's natural frequencies of pulsation. We shall return to those frequencies, the bell-like tones of a black hole, in Chapter 10.

When a rapidly spinning automobile wheel is slightly out of alignment, it can begin to vibrate, and its vibrations can begin to extract energy from the spin and use that energy to grow stronger and stronger. The vibrations can grow so strong, in fact, that in extreme cases they can even tear the wheel off the car. Physicists describe this by the phrase "the wheel's vibrations are unstable." Bill Press was aware of this and of an analogous behavior of spinning stars, so it was natural for him to ask, when he discovered that black holes can pulsate, "If a black hole spins rapidly, will its pulsations be unstable? Will they extract energy from the hole's spin and use that energy to grow stronger and stronger, and can the pulsations grow so strong that they tear the hole apart?" Chandrasekhar (who was not yet